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Personality traits predict the effects of Internet and academic self-efficacy on practical performance anxiety in online learning under the COVID-19 lockdown

Jon-Chao Hong[a,b], Wei Cao[c], Xiaohong Liu[c], Kai-Hsin Tai[a,b] and Li Zhao[c]

[a]Institute for Research Excellence in Learning Sciences, National Taiwan Normal University, Taipei City, Taiwan; [b]Department of Industrial Education, National Taiwan Normal University, Taipei City, Taiwan; [c]School of Education Science, Nanjing Normal University, Nanjing, China

ABSTRACT
Due to COVID-19, the primary teaching method has changed from traditional face-to-face teaching to online teaching. The present study explored the correlates between two personality traits, Neuroticism and Extraversion, and two types of self-efficacy, Internet self-efficacy and academic self-efficacy, on practical performance anxiety. Data from 273 technical college students were collected. Structural equation modeling analysis was performed. Results show that Neuroticism and Extraversion can predict students' practical performance anxiety through Internet and academic self-efficacy. Moreover, Neuroticism can negatively predict Internet and academic self-efficacy. Extraversion can positively predict Internet and academic self-efficacy. The two types of self-efficacy can positively predict practical performance anxiety. According to the results, it seems necessary to reduce students' practical performance anxiety by paying attention to their personality features and their self-efficacy.

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Personality; Internet self-efficacy; academic self-efficacy; practical performance anxiety; online learning; structural equation modeling

Introduction

During the COVID-19 pandemic, many schools responded to the call for "suspended classes, ongoing learning," and adopted online teaching to ensure that students could continue learning. Studies have shown that the use of online teaching methods affects students' cognitive-affective factors, which extends to their learning effect (Moreno, 2006). With the transition from the face-to-face to the online learning environment, some inexperienced students may lack confidence and experience negative emotions during online learning (Abdous, 2019). However, some students are full of confidence, believing that they can quickly adapt to this method and actively deal with all the problems encountered, thus showing positive emotions. Facing the same situation, why do different students behave differently? Trait activation theory (TAT), a personality theory of task functioning that integrates trait with situationism (Tett & Burnett, 2003; Tett et al., 2013), can explain this phenomenon. TAT proposes a central assumption that participants show consistency in their thoughts and actions, originating from a more stable personality trait (Scheuble et al., 2019). The model of the personality is often used to predict human attitudes and behaviors in many empirical studies (Costa & McCrae, 1992; Mount & Barrick, 1995). Considering this, this study takes personality traits as a factor affecting students' emotions and behaviors in online learning.

CONTACT Li Zhao li.zhao@njnu.edu.cn School of Education Science, Nanjing Normal University, 122 Ninghai Road, Nanjing, Jiangsu, 210097, China

Studies have shown that unexpected events cause behavioral adaptation. In the face of the same unexpected event, different levels of self-efficacy will lead to different adaptive behavior (Schipper et al., 2018). For example, a high level of self-efficacy promotes online learning interaction, and improves students' confidence and autonomy in online learning (Prior et al., 2016). Conversely, a sense of unease is often exacerbated in online learning by the students' lack of confidence in their academic and technical performance, leading them to experience feelings of anxiety (Hedges, 2017). Therefore, self-efficacy is also one of the factors that affect students' emotions and behaviors in online learning.

However, the level of students' self-efficacy and anxiety may vary from person to person. For college students in majors with strong practical tasks (i.e., tasks which require students to practice by themselves, and which involve many complicated steps), reducing practical performance anxiety and successfully completing practical tasks is very important to cultivate their vocational skills. Therefore, exploring the relationship between personality traits, self-efficacy, and practical performance anxiety in the online learning of college students will provide a new research reference for college students' better practical task performance. Based on this, a conceptual framework which indicates that personality can predict performance anxiety mediated by self-efficacy was formed in this study. The findings of this study provide a series of references related to the practical performance anxiety of college students in online practical courses during the current pandemic and its subsequent waves. They also provide important insights into the development of effective mental health interventions.

Literature review

Neurotic and extrovert personalities

Personality refers to individual differences in thinking, feeling and behavior patterns (Kazdin, 2000). Gray (1970) proposed a biological model of personality which comprises the behavioral inhibition system (BIS) and the behavioral approach system (BAS). An individual can prompt their emotions through the bilateral traits of BIS and BAS, a number of different personality traits have been suggested for inclusion in the BIS and BAS. Accordingly, researcher proposed that Neuroticism and Extraversion can be examined in mood-related studies, respectively (Smillie et al., 2006). The BIS and BAS operate within the same conceptual space as Neuroticism and Extraversion traits (Eysenck, 1970). Therefore, two personality traits, Neuroticism and Extraversion, were adopted in this study to explore their effects on online learning.

Because personality is a set of patterns of behavior and thought that are habitual, it can be used to manage different kinds of challenges in learning (Tavitiyaman et al., 2021). While the role of personality during the COVID-19 crisis is not yet fully understood, it is assumed that personality plays a key role in handling a pandemic situation. Studies have shown that individuals with certain personality traits with high levels of optimism tend to handle these emotions better (Bartone et al., 2018). The study analyzes how individuals react to the coronavirus pandemic based on their two types of personality traits.

Internet and academic self-efficacies

Self-efficacy has been defined as the belief one has in one's ability to successfully complete a task (Bandura, 1977). When relating self-efficacy to online learning, researchers have approached it from different perspectives (Hodges, 2008; Zhao, Liu, et al., 2021). As the concept of transaction involves interaction among the environment, individuals, and the structure, the framework of interactivity in the context of distance education comprises two interactivities, one is between the learner and the environment, and the other is between the learner and the content (Moore, 2013). Accordingly, considering learners' interaction ability and their confidence in successfully performing online learning, two types of self-efficacy, Internet self-efficacy (ISE) (i.e., learner-online system interaction) and academic self-efficacy (ASE) (i.e., learner-content interaction), were taken into account in this study.

ISE reflects a person's confidence in interacting with an information technology system. It may be especially important for online students who need certain human-computer interaction skills to learn smoothly (Kuo et al., 2014). Low ISE may prevent many students from participating in learning activities. This could result in dissatisfaction with online learning or anxiety. (Liang & Tsai, 2008; Tsai, 2012; Tsai et al., 2011). A student's confidence in his or her ability to complete academic tasks is known as ASE. This is a part of the self-efficacy domain (Schunk & Pajares, 2002). Moreover, ASE and ISE are relevant self-efficacy constructs in online learning (Joo et al., 2000; Kuo & Belland, 2019; Torkzadeh et al., 2006). However, the psychological stability and behavioral changes of an individual can also be affected by the sudden changes in their environment (Donovan & Rossiter, 1982). The stimulus is the external forces that influence the individual's psychological state. The crisis stimulus of the COVID-19 pandemic affects students' internal psychological processes (Pandita et al., 2021). As a result, students are experiencing stress due to their inability to practice by themselves and to barriers to learning in practical courses.

One's response efficacy is related to his or her psychological adaptation to predict one's behavior (Bradley et al., 2020). Self-efficacy is a manifestation of psychological changes. Studies have shown that personality traits play an important role in self-efficacy (Özdemir et al., 2019; San-Martin et al., 2020). When students with different personality traits interact with the transactional media (functional interaction, calling up various functions on a site, such as instructional video downloads) and transactional content (cognitive interaction, interaction on learning content, such as group discussion questions), their psychology will also change, which will affect their practical performance. Therefore, this study explores how neurotic and extrovert students change ISE and ASE when they interact with their online learning in the crisis.

Practical performance anxiety

Manual performance is a critical element in practical education, in which quality and safety are important concerns and are needed to meet professional standards (Al-Ghareeb et al., 2019; Benner et al., 2010). Professional standards and requirements of manual performance bring stress to learners. Therefore, learning practical skills (e.g., nursing skills) can cause anxiety for learners who aim to master the manual skills (Najjar et al., 2015; Shearer, 2016). Studies have found that online students are more anxious than traditional students in practical courses (e.g., data-process-oriented courses), such as courses on statistics and research methods (Hedges, 2017; Rapp-McCall & Anyikwa, 2016). During the pandemic, the sudden shift from the offline to the online teaching mode has created personal adjustment burdens. Furthermore, online learning restricts practical courses in terms of location and tools, and students cannot operate on-site. This has brought great anxiety to students, and this anxiety will affect their practical performance in practical courses.

According to Huberty (2009), anxiety is a normal emotional state which may present in a variety of situations, one being learning. The Processing Efficiency Theory (PET) addresses anxiety significantly affects performance (Eysenck et al., 2007). According to PET, when individuals think that one of their goals is being threatened, they will identify the source of the threat. This will have a negative effect on the individual's performance (Eysenck et al., 2007). During the COVID-19 outbreak, learning settings suddenly changed so that learners could not experience real practice, causing disruptions in their learning process and creating some kinds of anxiety (Bao, 2020; Tavitiyaman et al., 2021). Therefore, this study explored practical performance anxiety (PPA) as the college students' emotional state of anxiety, caused in the online data-process-oriented practical courses. Based on this, this study proposes two research questions: (1) What is the difference between the level of self-efficacy of Neuroticism and Extraversion?

and (2) How do their different levels of self-efficacy affect students' practical performance anxiety in practical courses?

Research model and hypotheses

Research model

A mind-set reflects one's efficacy that is not fixed, but is changeable through environmental changes (Smith & Capuzzi, 2019). Accordingly, based on the two research questions, this study drew on TAT to explore the structural relationships between Neuroticism, Extraversion, ISE, ASE, and practical performance anxiety in college students' online learning during COVID-19. The structural research model is described as Figure 1.

Hypothesis

Studies have shown that Neuroticism has a negative effect on self-efficacy (Correa et al., 2013; Judge et al., 2007; Pocnet et al., 2017). On the contrary, Extraversion has been shown to be positively related to self-efficacy (Judge et al., 2007; Saleem et al., 2011). Moreover, individuals who score highly on Extraversion tend to use the Internet more frequently (Mark & Ganzach, 2014) and are quicker to adapt to change (Pornsakulvanich et al., 2012; Walczuch & Lundgren, 2004). Given the prevalence of online learning in the COVID-19 lockdown, having a good understanding of how one's personality may drive self-efficacy is of importance in promoting online learning. Therefore, how the two types of personality relate to the two types of self-efficacy is hypothesized as follows:

H1 Neuroticism is negatively related to students' ISE in online learning.

H2 Neuroticism is negatively related to students' ASE in online learning.

H3 Extraversion is positively related to students' ISE in online learning.

H4 Extraversion is positively related to students' ASE in online learning.

According to PET, factors that might increase anxiety include expecting "deficiency" (Paige & Morin, 2015), and the level of skill fidelity when students face training (Dzioba et al., 2014). Studies have shown that an increase in self-efficacy has a positive impact on students' professional efficiency to tackle new tasks, and management of academic stressors (Alghamdi et al., 2020; Salazar & Hayward, 2018). Students with low self-efficacy have inferior performance, lack confidence in a range of relevant tasks, and show anxiety (Heckel & Ringeisen, 2019). A recent study also showed that high self-efficacy may decrease anxiety in online learning environments (Heckel & Ringeisen, 2019; Huang & Mayer, 2019). Low self-efficacy is a potential source of

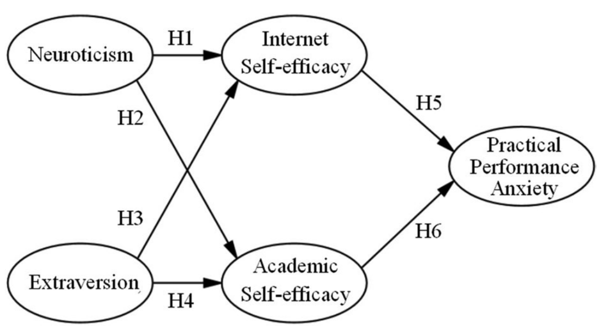

Figure 1. Research model.

anxiety, especially for online students (Abdous, 2019). However, there is as yet limited evidence of how college students' practical performance anxiety is impacted by the two types of self-efficacies in online learning during COVID-19; thus, the relationship between the two types of self-efficacy and practical performance anxiety was hypothesized as follows:

H5: ISE is negatively related to students' PPA.

H6: ASE is negatively related to students' PPA.

Fu et al. (2021) highlighted that internal processes of learning usually mediate the relationship between individual traits and emotion response. It has been shown that there is a positive correlation between Neuroticism and negative emotion, whereas extraversion was found to have a positive association with positive emotion (Gomez & Gomez, 2002). In searching for the relationship between different types of personality and anxiety, Tavitiyaman et al. (2021) found that students with high levels of neuroticism can predict high levels of anxiety in learning with technology; but to those students who with high levels of extraversion can experience low levels of performance anxiety. However, few studies have been done to understand how personality traits can enhance students' perceived self-efficacies in online data-process-oriented learning, and consequently how they affect their performance anxiety during COVID-19. The following hypotheses were thus proposed:

H7: Neuroticism is significantly related to PPA mediated by ISE.

H8: Neuroticism is significantly related to PPA mediated by ASE.

H9: Extraversion is significantly related to PPA mediated by ISE.

H10: Extraversion is significantly related to PPA mediated by ASE.

Method

Participants and data collection

This study adopted the random and snowball sampling methods. From April 20 to 30, 2020, students majoring in Electronic and Information Engineering from different universities in Jiangsu province, China were surveyed. Firstly, questionnaires were posted on a survey Website, then notification was sent to those association chairmen of vocational college students, asking them to post the survey questionnaire to their WeChat groups. In order to ensure confidentiality, this study adopted an anonymous questionnaire survey.

A total of 345 questionnaires were returned, and after preliminary screening of invalid data, 273 valid questionnaires were obtained. The response rate of the questionnaire was 79.1%, which met the requirement of building an adequate model. When models contain up to five constructs, and more than three items should be included to have item communalities of 0.6 or higher as sample size over 100 (Hair et al., 2019). The 273 usable return data met the minimum sample size requirement.

Respondents

Table 1 summarizes the respondents' gender, age, time spent on online courses, and time spent on data-processing online courses.

The online course of this research

According to the Dictionary of Occupational Titles, workers function in a job by interacting with people, data, and things (Cain & Treiman, 1981). Similarly, students in learning also

Table 1. Respondent profiles (N = 273).

Profile	Item	Frequency	Percentage (%)
Gender	Male	77	28.2%
	Female	196	71.8%
Age	17-19	179	65.6%
	20-22	91	33.3%
	≥23	3	1.1%
Time spent on online courses	Less than 2 h/weeks	27	9.9%
	2h-4h/ week	137	50.2%
	4h-6h/week	72	26.4%
	More than 6 h/week	37	13.6%
Time spent on data-process-oriented online courses	Less than 1 h/week	52	19.0%
	2h-3h/week	203	74.4%
	4h-5h/week	15	5.5%
	More than 5 h/week	3	1.1%

play a role by interacting with people, data, and things. Data-process-orientation refers to interacting with data, such as analyzing data, designing data, capturing data, etc. Thing-process-orientation refers to interaction with objects, such as analyzing objects, designing objects, producing objects, etc. Data-process-oriented courses mainly allow students to learn some software to master the method of processing data, such as in a Digital Image Processing course. On the other hand, thing-process-oriented courses mainly allow students to operate some equipment to process products, such as in a Fundamentals of CNC Machine Tool Operation course.

Different course characteristics require different implementation conditions. For example, courses that are data-process-oriented should be easier to conduct via online learning, whereas thing-process-oriented courses which involve high-level psychomotor skills (i.e., manual work; Alqudah et al., 2020) may be better promoted by way of blended learning (Lapitan et al., 2021; Zhao, He, et al., 2021). Blended learning aims to combine the advantages of traditional learning with those of online learning. During the COVID-19 lockdown, many schools only adopted online learning to ensure that students could continue learning. It is difficult to implement blended learning (Meulenbroeks, 2020). Based on the consideration of the relationship between course characteristics and teaching conditions, it is therefore worth focusing on data-process-oriented rather than thing-process-oriented courses. Thus, this study selected a data-process-oriented course for college students as its focus during the COVID-19 pandemic.

Most courses in Electronic and Information Engineering are mainly based on training students' practical performance. For example, the course of Fundamentals of Engineering Drawing requires students to understand and master the drawing method of dimension labeling and configuration of three-dimensional and composite bodies, and the expression method of machine parts. They should also be able to understand and draw engineering drawings which can be considered as data-process-oriented work.

First of all, the teacher carefully prepares online teaching work, including making online courseware, self-compiled electronic teaching plans, electronic teaching materials for textbooks, self-compiled experimental instructions, etc. Secondly, in the process of online teaching, in order to ensure the effectiveness of online teaching, the teacher adopts a combination of course recording and live broadcasting. Important operating videos are recorded and broadcast, while pre-class and in-class teaching are conducted in live broadcast mode. When students encounter various problems in their practice, they will ask questions through screenshots in the QQ course group. The teacher patiently answers through the video, and points out the students' operation problems. In addition, students can also answer and help each other. Finally, the teacher arranges related tasks, such as drawing various views of gears, springs, and other mechanical parts. Students use AutoCAD software offline to complete homework in individual or group form.

Instrument

The development of the questionnaire

In order to test the proposed hypotheses, the study used items developed in prior studies to measure the constructs. A team of experts specializing in self-efficacy studies review the original set of items. Based on their feedback, an expanded set of six items for each type of personality: Neuroticism and Extraversion; six items for each type of self-efficacy; and seven items for PPA were developed. All of these scales used a 5-point Likert rating (5 = *strongly agree*, 4 = *agree*, 3 = *neither agree nor disagree*, 2 = *disagree* and 1 = *strongly disagree*). The first-order confirmatory factor analysis was then applied to delete any unsuitable items. Moreover, the reliability and validity of the constructs were retested as shown in Table 2.

Measurement

Personality: According to McCrae and Costa (1987), the five-factor model is a universal and comprehensive framework for describing individual personality differences. We selected items related to Neuroticism and Extraversion from the International Personality Item Pool (Donnellan et al., 2006) to design the two types of personality to measure individual differences in traits. The Neuroticism scale contains items such as, "I can't calm down under tension" and "Sometimes I tighten my nerves and suffer from tension." The Extraversion scale contains items such as, "I will take the initiative to talk to people" and "I am willing to help others."

Self-efficacy: The two types of self-efficacy are a response to the psychometric challenges to the Self-Efficacy Scale (SES; Sherer et al., 1982). The ISE scale includes items such as, "If there is a new human-computer interaction problem in online learning, I can solve it by myself" and "For me, the problem of human-computer interaction encountered in online learning will be overcome as long as it takes more time." The ASE scale includes items such as, "In online learning, I have the ability to understand and solve problems about new learning content" and "In online learning, when I encounter difficult content, I will try to overcome it."

Practical performance anxiety: The anxiety shown when a person is facing fear is considered measurable because they have the same physical manifestations. The anxiety scale has been shown to be useful for self-reporting of fear and for subjects with limited capacity (Gustad et al., 2005). Adapted from Gustad et al. (2005), items of the PPA include, "The teacher's online demonstration operation steps are unrealistic, so I am still afraid of making mistakes" and "Even if the online teacher has fully demonstrated the operation steps, I still worry about not doing well."

Results

Questionnaire survey and quantitative analysis methods were adopted in this study, moreover, the findings are presented through structural equation modeling (SEM). That is, data collected were analyzed with the SPSS 21 and AMOS 26 software. The results are as follows.

Item analysis

In the beginning, the original questionnaire had 31 items in total, including six items for Neuroticism, six for Extraversion, six for ISE (ISE), six for ASE (ASE), and seven for the PPA construct. Firstly, when factor loadings are below 0.4, items were deleted. (Hair et al.,

Table 2. First-order CFA analysis.

Fit indices	χ^2/df	GFI	AGFI	NFI	CFI	RMSEA
Threshold values	<3.00	>0.90	>0.90	>0.90	>0.90	<0.80
Result values	1.152	0.957	0.935	0.971	0.996	0.024

2017). Secondly, the first-order confirmatory factor analysis (CFA) was applied to test item suitability by deleting those items with the highest residual values in each construct. Items with a residual value over 0.5 were also deleted until those values of first-order CFA met the thresholds suggested by Hair et al. (2019) (see Table 2). Finally, 15 remaining items were kept for the further analysis, including three items each for Neuroticism, Extraversion, ISE, ASE, and PPA.

Reliability and validity analysis

Reliability is verified by internal consistency reliability (Cronbach's α) and composite reliability (CR). As shown in Table 3, the Cronbach's α values all exceeded .800, and the CR for all constructs was greater than 0.802 (ranging from 0.802 to 0.950) (Hair et al., 2019), showing that the study constructs had acceptable reliability.

Convergent validity can be tested by examining the average variance extracted (AVE) and factor loading which values should be higher than 0.5. (Hair et al., 2019). Table 3 shows that all AVE values are above 0.57 and FL are 0.86. Therefore, the convergent validity of the model was accepted.

Construct discriminant validity (CDV) can be achieved when the square root of AVE over the Pearson correlation values between constructs (Fornell & Larcker, 1981). Table 4 shows that all square roots of the AVEs (as shown on the diagonal) were greater than Pearson correlation in the off-diagonal constructs. The CDV was therefore considered to be acceptable.

Model fit analysis

AMOS was applied to examine the fit indices and tested path analysis of the structural model. As shown in Table 5, the indices exhibited perfect fitness in terms of the structural model

Table 3. Reliability and validity analysis.

Construct	Item	*M*	*SD*	FL	CR	AVE	Cronbach's α
Neuroticism	N2	2.66	1.24	0.87	0.86	0.68	0.85
	N3	2.72	1.23	0.92			
	N4	2.96	1.25	0.66			
Extraversion	E2	3.46	1.17	0.77	0.80	0.57	0.80
	E3	4.08	0.97	0.77			
	E5	3.37	1.11	0.75			
ISE	ISE2	3.97	0.96	0.90	0.94	0.85	0.94
	ISE4	3.99	0.96	0.94			
	ISE5	4.03	0.96	0.91			
ASE	ASE1	4.22	0.81	0.90	0.92	0.79	0.92
	ASE3	4.21	0.83	0.91			
	ASE5	4.20	0.86	0.85			
PPA	PPA3	2.69	1.13	0.88	0.95	0.86	0.95
	PPA4	2.75	1.13	0.96			
	PPA6	2.75	1.15	0.95			

Note.
$^{***}p < .001$.

Table 4. Discriminant validity analysis.

Constructs	Neuroticism	Extraversion	ISE	ASE	PPA
Neuroticism	**0.823**				
Extraversion	0.000	**0.758**			
ISE	0.314	0.264	**0.919**		
ASE	0.227	0.355	0.165	**0.887**	
PPA	0.183	0.219	0.359	0.448	**0.929**

Table 5. Fit indices of the structural model.

Fit indices	χ^2/df	GFI	AGFI	NFI	CFI	RMSEA
Threshold	<3.00	>0.90	>0.90	>0.90	>0.90	<0.08
Result values	1.771	0.932	0.903	0.953	0.979	0.053

proposed in this study (χ^2 = 148.760, $df = 84$, $\chi^2/df = 1.771$, GFI = 0.932, AGFI = 0.903, NFI = 0.953, CFI = 0.979, RMSEA = 0.053).

Path analysis

The standardized path coefficient from Neuroticism to ISE (β = −0.314, $p < 0.001$; H1 supported) was significant. The path from Neuroticism to ASE (β = −0.227, p < 0.001; H2 supported) was also significant. The path from Extraversion to ISE (β = 0.264, $p < 0.001$; H3 supported) was significant, as was the path from Extraversion to ASE (β = 0.355, $p < 0.001$; H4 supported) and the path from ISE to PPA (β = −0.293, $p < 0.001$; H5 supported). The path from ASE to PPA (β = −0.399, $p < 0.001$; H6 supported) was also significant.

The explanative power (R^2) and effect size (f^2) are the measures of predictability in structural models (Hair et al., 2019). The R^2 values range from 0 to 1. The effect size f^2 of 0.02 is small, 0.15 is moderate, and 0.35 is substantial (Hair et al., 2019). As shown in Figure 2, the explanatory power of Neuroticism and Neuroticism on ISE was 17%, with an effect size f^2 of 0.20; the explanatory power of Neuroticism and Neuroticism on ASE was 18%, with an effect size f^2 of 0.22; the explanatory power of ISE and ASE on PPA was 28%, with an effect size f^2 of 0.39. All statistical values were thus acceptable for this study (See Figure 2).

Indirect effect

We conducted 5,000 resample bootstrapping with 95% bias-corrected confidence intervals to provide an evidence of the significance of the indirect effect. Table 6 shows that the bootstrapping confidence intervals' lower and upper bounds of indirect effects did not include zero, indicating all of the indirect effects in the structural equation model were significant. Conclusively, Neurotic and Extrovert personalities had an effect on their practical performance anxiety through the mediation of Internet and academic self-efficacies according to the bootstrapping results.

Discussion

Drawn on TAT, this study explored the structural relationships between Neuroticism, Extraversion, ISE, ASE, and practical performance anxiety in college students' online learning during COVID-19.

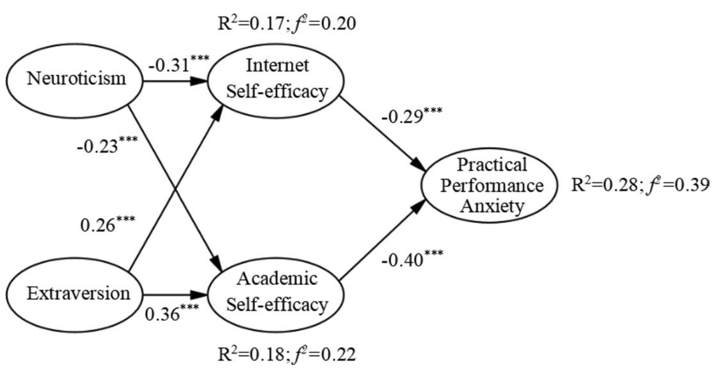

Figure 2. Verification of the research model.

Table 6. Indirect effect analysis.

Indirect effect	Estimate	SE	bias-corrected 95%CI Lower	bias-corrected 95%CI Upper
Neuroticism → ISE → PPA	0.111	0.044	0.038	0.206
Neuroticism → ASE → PPA	0.109	0.047	0.036	0.22
Extraversion → ISE → PPA	−0.093	0.04	−0.191	−0.032
Extraversion → ASE → PPA	−0.17	0.049	−0.292	−0.093

To this aim, the present study provided conceptual and empirical evidence that the relationship between personality and practice performance anxiety, the study results will be further discussed based on the questions.

Individual behavior can be predicted by personality traits which can also explain differences in individual behavior in similar situations (Watjatrakul, 2020). When faced with a new online learning environment, students with higher Neuroticism suffer more pressure and react to negative situations than those who are more emotionally stable (Vinograd et al., 2020). Therefore, they may be more prone to having lower self-efficacy and anxiety during human-computer interaction and learning. On the contrary, extroverted people will feel more positive affect with high activation by bringing their enthusiasm to online learning, and when faced with difficulties, they are more likely to respond positively (Spark & O'Connor, 2020). The present study further confirms this point that Neuroticism has a negative effect on ISE and ASE, while Extraversion has a positive effect on ISE and ASE. This indicates that H1 and H2 were negatively supported, while H3 and H4 were positively supported.

For college students in majors with strong practical tasks, practical task performance plays an important role in the development of vocational skills. Self-efficacy regulates the relationship between task importance and task anxiety (Alhadabi & Karpinski, 2020). The benefit of high self-efficacy is that it reduces anxiety and eases students' stress (Zhou et al., 2020). ASE is another positive predictor of students' participation in online learning (Yu et al., 2020). In addition, ISE is a positive antecedent of students' participation in the online learning environment (Kuo et al., 2014). Students with a high level of online learning self-efficacy have been found to be more interested in mastering technology and in learning-related challenges, and tend to experience lower levels of anxiety during the online learning process (Abdous, 2019). The present study further confirms this point. The results show that students' ISE and ASE had negative effects on their PPA. This indicates that H5 and H6 were negatively supported.

In examining H7-H10, the result indicated that the two types of self-efficacy played the mediating role between personality type and performance anxiety in practical skill development. As expected, positive association between Neuroticism and negative emotion, and a positive correlation between extraversion and positive emotion (Gomez & Gomez, 2002); the present study revealed that Neuroticism was positively related to performance anxiety, while Extraversion was negatively related to it mediated by two types of self-efficacy in online data-process-orientation during COVID-19.

Conclusion

By adapting TAT, the study provides some contributions to expand our understanding of the complex relationships surrounding personality, self-efficacy, and practice performance anxiety in online learning during COVID-19. This study provides evidence that college students' PPA in online learning can be decreased when their ISE and ASE are increased. In addition, the finding shows that Extraversion positively affects ISE and ASE, whereas Neuroticism negatively affects ISE and ASE.

Implications

Facing online learning, the anxiety of college students is not optimistic. Thus, reducing neuroticism may reduce future psychopathology and improve the quality of life in the present and future. Finding out the relationship between them will provide a new perspective for effectively solving the practical performance anxiety of students in online practical courses.

The theoretical contribution of this study is that it fills the gap in the PPA literature which has paid limited attention to the effects of individual differences and self-efficacy on college education in online learning. That is, this study first proposes a new framework for examining and successfully confirming the relationships between two types of self-efficacy, namely Internet self-efficacy and academic self-efficacy, and PPA in online learning.

The practical contribution of this study is that the results show that types of personality and self-efficacy differentially significantly influence their anxiety in practice manipulation courses with instruction under COVID-19. Therefore, this study makes the following suggestions: Firstly, the neurotic groups in personality traits are more sensitive and are prone to various mental health problems. However, there is grouping evidence that personality is malleable (MacLean et al., 2011). Therefore, the teacher can guide students in online mindfulness training, and guide them to improve their self-efficacy through enhancing their ability in Internet functional interactivity and content interactivity.

Secondly, human-computer interaction skills training is an important way to improve Internet self-efficacy (Torkzadeh & Dyke, 2002). Individuals who receive more support and guidance in online learning seem to be more likely to have a positive attitude toward the Internet. Their self-confidence in online learning is significantly improved (Wu & Tsai, 2006). Therefore, before online teaching, schools and teachers need to carry out human-computer interaction skills training for college students, explain the operating procedures and precautions in the human-computer interaction process, and help students familiarize themselves with the online course learning environment, resource usage methods, interactive communication methods, etc. Students' information literacy and Internet self-efficacy are effectively improved through the efforts of schools and teachers.

Limitations and future study

Although the study provides some important contributions to the literature, it has some limitations that should be addressed in future studies. First, the data of this study were collected in one province of China through the snowball sampling method, and therefore cannot represent all college students in China. More and larger representative samples will be needed in the future to assess the extent to which these findings are applicable to other population groups and other countries to confirm the assumptions of this study. Second, the study was focused on a data-process-oriented online course to examine participants' PPA. Future studies may compare data-process-oriented courses to thing-process-oriented courses, such as woodworking for making furniture, to examine the PPA. Third, technical and vocational students may experience more anxiety than general college students in the same course due to stereotype reasoning regarding their poorer academic performance before studying college courses in China (Guo & Wang, 2020). Finally, there were too many female respondents in this study, which may have affected the results of the experiment. Future studies should aim to recruit a balanced proportion of male and female participants. Thus, it would be interesting to compare the same course, such as AutoCAD, for general college students and for technical college students with those constructs in this study.

Notes on contributors

Jon-Chao Hong is a professor in the Department of Industrial Education at National Taiwan Normal University and also a member of the Institute for Research Excellence in Learning Science.

Wei Cao is a teacher in School of Loufeng, Suzhou Industrial Park, Suzhou, China. She is mainly engaged in ICT course instructional design, and teaching strategies. She holds an M.Ed. in Educational Technology from Nanjing Normal University (NNU) and a B.Ed. in Educational Technology from Yangzhou University.

Xiaohong Liu is a doctoral candidate at School of Educational Science, Nanjing Normal University (NNU). She is mainly engaged in online learning, teaching strategies, and technology application in education. She holds an M.S. in Educational Technology from Nanjing Normal University, and a B.Ed. in Educational Technology from Jiangxi Normal University. Kai-Hsin Tai is a post-doctoral fellow of National Taiwan Normal University and also a member of the Institute for Research Excellence in Learning Science.

Li Zhao is an associate professor in School of Education Science, Nanjing Normal University, Nanjing, China. She is mainly engaged in online learning and technology application in education. She holds a Ph.D. in Educational Technology from Nanjing Normal University (NNU), and an M.S. and a B.Ed. in Educational Technology from Nanjing Normal University (NNU), and also completed a master's program of science in Instructional Technology at The University of Southern Mississippi (USM). Presently, she serves as an Editor of Cogent Education.

ORCID

Jon-Chao Hong http://orcid.org/0000-0003-3120-5861
Xiaohong Liu http://orcid.org/0000-0003-4931-8370
Kai-Hsin Tai http://orcid.org/0000-0002-7786-7329
Li Zhao http://orcid.org/0000-0003-1000-1296

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